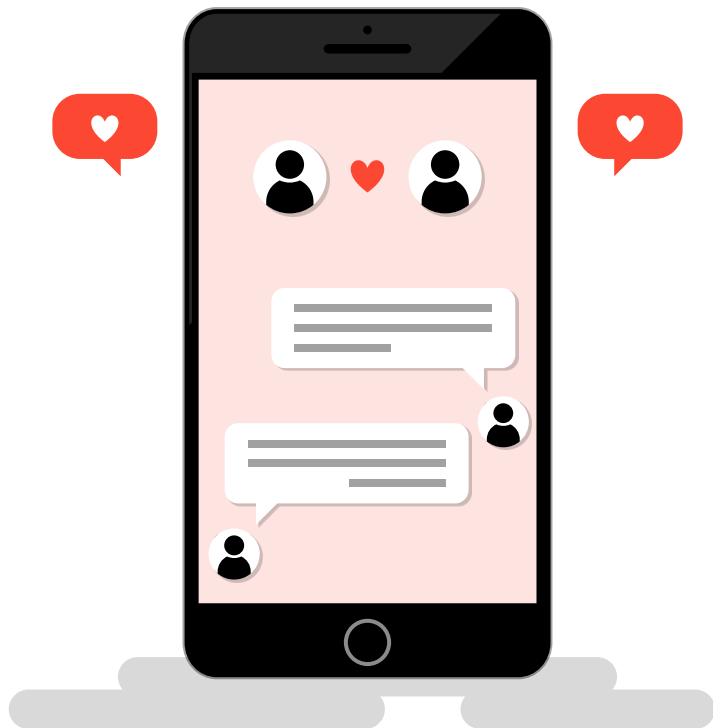




Tinder Experiment DATASCI 241

Aryana Far, Jonah Grossman, Jun Park, Derek Yao



Research Question

What is the effect of graduate-level higher education on women's overall likability in American heterosexual online dating contexts, both real or simulated?



Hypothesis

Two-tailed Hypothesis: Graduate-level education has some effect on women's likability

Existing literature is contradictory:

- **Positive effect:** intelligence, ambition, potential earnings
 - ([Eika et al., 2017](#), [De Hauw et al., 2017](#))
- **Negative effect:** social biases, intimidation, compatibility
 - ([Egebark et al., 2021](#), [Neyt et al., 2023](#))



Failure...Two Experiments

Experiment A

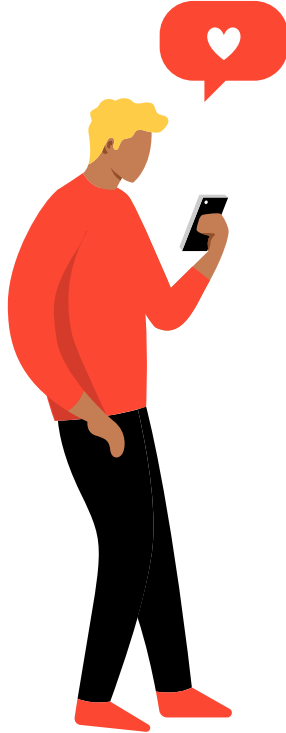
field experiment
measures actual likability
1800 observations
no covariates

Experiment B

survey experiment
measures stated likability
200 observations
covariates



What is “likability”?



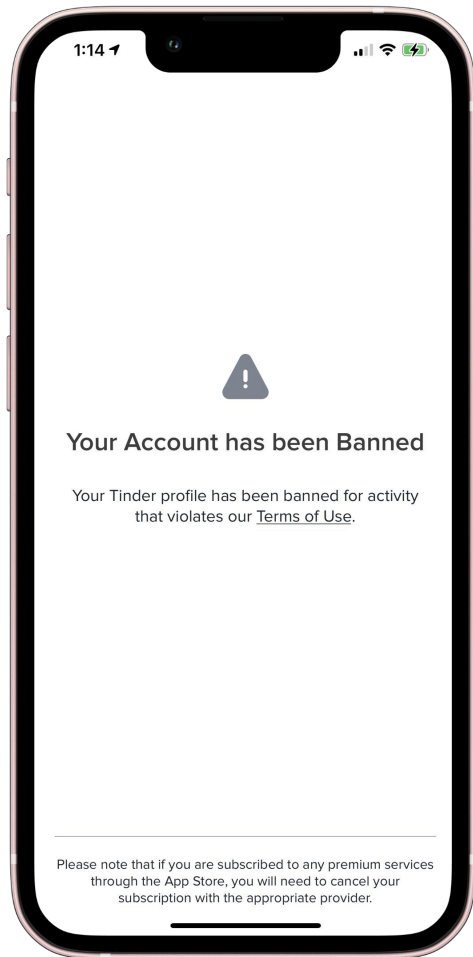
Tinder

- actual degree of appeal that other users perceive
- actually “Swiped Right” (liked)
- actually messaged

Survey

- stated degree of appeal that respondents perceive
- attractiveness rating
- stated they would “Swipe Right”
- stated they would message





Experiment A

Fictitious Profile Creation

- sourcing stock image
- 85% failure (attempted 24 accounts)
- 6 profiles each sent 300 likes→1800 observations

Profile Information

- women aged 25; 24-32 age range; strictly men
- 100 mile radius (strictly Bay Area)

Outcomes

- matches and messages

Treatment

- **treatment:** Master's or PhD
- **control:** High School, with generic service job

The Real Profiles



Profile A – Katrina

Bio: Waitress who loves chill matcha chats and photography!

Treatment: Not treated, high school



Profile B – Naomi

Bio: Currently a hostess at a local restaurant! I love to read and I love all animals!

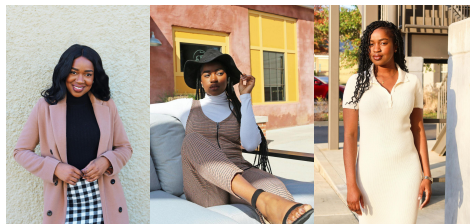
Treatment: Not treated, high school



Profile C – Samantha

Bio: Spreading joy, one dish at a time! Let's savor the flavors of life together!

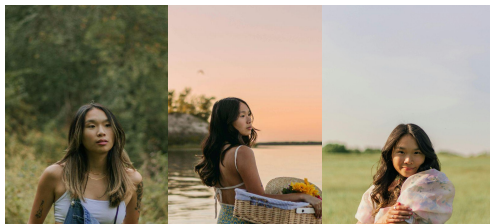
Treatment: Not treated, high school



Profile D – Aisha

Bio: Just earned my Master's at UC Berkeley – ready to conquer the world with a sprinkle of Berkeley spirit! Let's celebrate together!

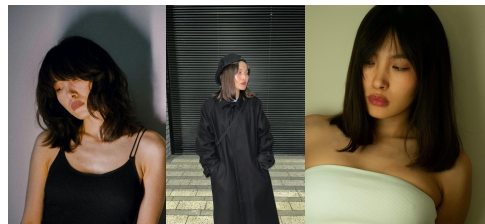
Treatment: Treated, Master's



Profile E – Jasmine

Bio: Stanford PhD geek with endless hobbies – let's chat and geek out together!

Treatment: Treated, PhD

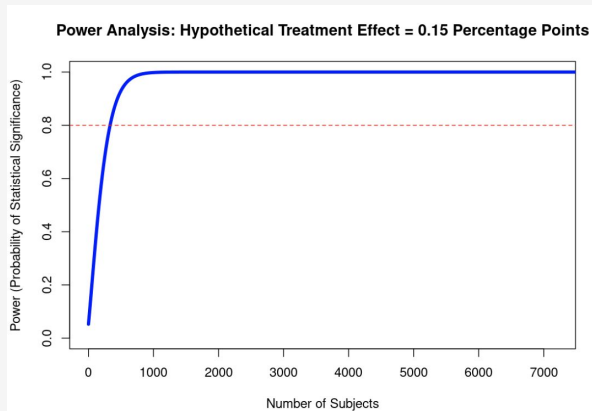


Profile F – Bella

Bio: Freshly minted with my Masters from UC Berkeley, armed with ambition and a hint of Berkeley's zest for life! Join me in celebrating our victories and conquering new horizons together!

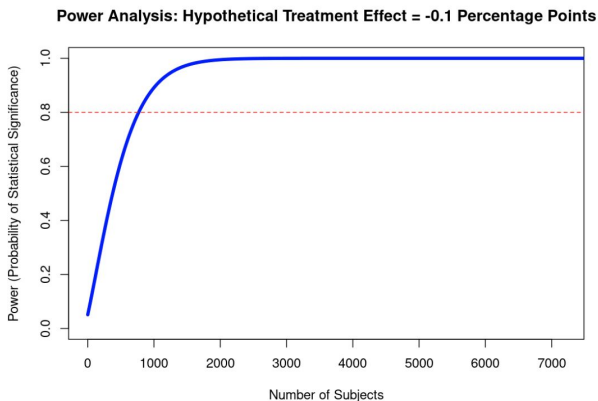
Treatment: Treated, Master's

Experiment A Power



Positive Treatment Effect

- The first graph shows the power analysis for a hypothetical positive treatment effect of 0.15 percentage points
- To achieve at least 80% power, we would to send out about 750 likes

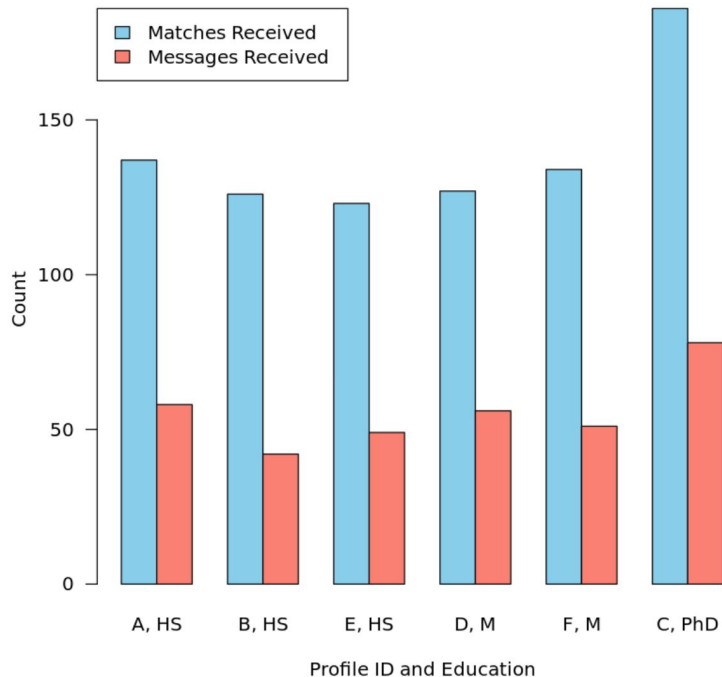


Negative Treatment Effect

- The second graph shows the power analysis for a hypothetical negative treatment effect of -0.1 percentage points
- To achieve at least 80% power, we would to send out about 1000 likes

Experiment A Results

Counts of Matches Received and Messages Received



Comparison of Models with Liked Outcome

	<i>Dependent variable:</i>	
	liked_back	
	(1)	(2)
treated	0.273*** (0.095)	-0.214* (0.116)
profile_idA		-0.174 (0.116)
profile_idB		-0.323*** (0.117)
profile_idC		0.704*** (0.166)
profile_idD		-0.095 (0.165)
profile_idE		-0.364*** (0.117)
profile_idF		
Constant	-0.286*** (0.067)	
Observations	1,800	1,800
Log Likelihood	-1,238.512	-1,223.818
Akaike Inf. Crit.	2,481.024	2,459.636
Note: * p<0.1; ** p<0.05; *** p<0.01		

Comparison of Models with Messaged Outcome

	<i>Dependent variable:</i>	
	messaged	
	(1)	(2)
treated	0.266** (0.122)	-1.586*** (0.154)
profile_idA		-1.428*** (0.146)
profile_idB		-1.815*** (0.166)
profile_idC		0.540*** (0.202)
profile_idD		0.114 (0.213)
profile_idE		-1.634*** (0.156)
profile_idF		
Constant	-1.617*** (0.090)	
Observations	1,800	1,800
Log Likelihood	-861.099	-855.431
Akaike Inf. Crit.	1,726.198	1,722.862
Note: * p<0.1; ** p<0.05; *** p<0.01		

Experiment B

Survey Creation

- **Platform:** Pollfish
- **Demographic:** Heterosexual males aged 18-34, various education levels and races
- **Profiles:** Two fictitious females, age 25, presented in similar attractiveness

Outcomes

- attractiveness rating
- stated "swipe right" behavior
- stated messaging behavior

Treatment

- Respondents reviewed 1 untreated (high school) and 1 treated (graduate) profile
- Total observations: 200 (100 respondents × 2 observations each)

Randomization Process

- There were 2 survey versions with swapped images to avoid bias
- Random 50/50 split for the two survey versions

The Simulated Profiles



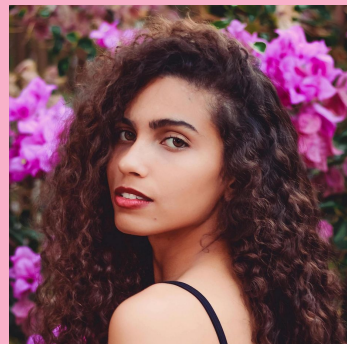
Woman A – Isabella

Looking for: a life partner

Highest education

obtained: high school diploma

Height: 5'3



Woman A – Isabella

Looking for: a life partner

Highest education

obtained: high school diploma

Height: 5'3



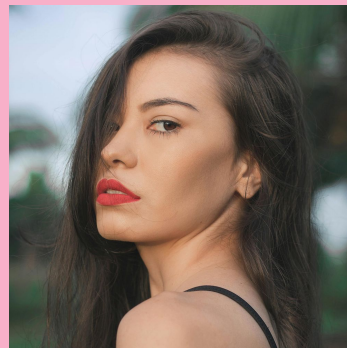
Woman B – Sofia

Looking for: a life partner

Highest education

obtained: graduate school degree

Height: 5'3



Woman B – Sofia

Looking for: a life partner

Highest education

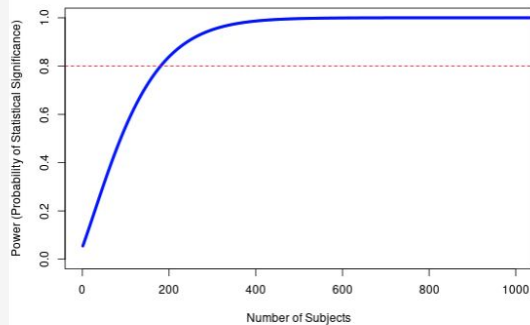
obtained: graduate school degree

Height: 5'3

Experiment B Power



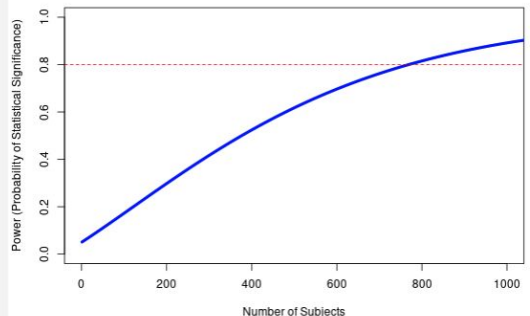
Power Analysis: Hypothetical Treatment Effect = 0.2 Percentage Points



Positive Treatment Effect

- The first graph shows the power analysis for a hypothetical positive treatment effect of 0.2 percentage points
- To achieve at least 80% power, a study would need to include around 181 subjects.

Power Analysis: Hypothetical Treatment Effect = -0.1 Percentage Points

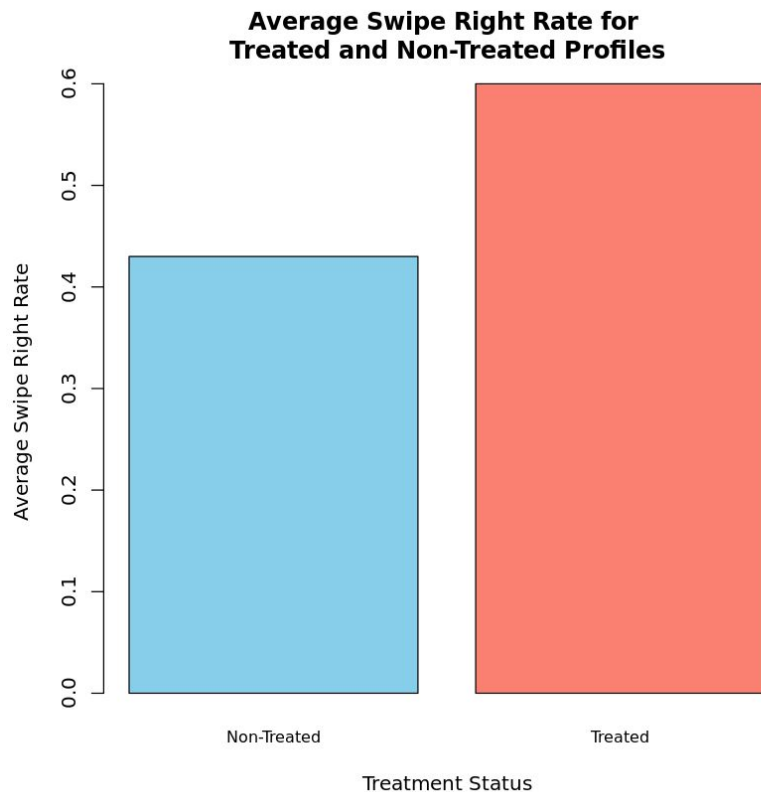


Negative Treatment Effect

- The second graph shows the power analysis for a hypothetical negative treatment effect of -0.1 percentage points
- To achieve at least 80% power, a study would need to include around 770 subjects.



Experiment B Logistic Regression



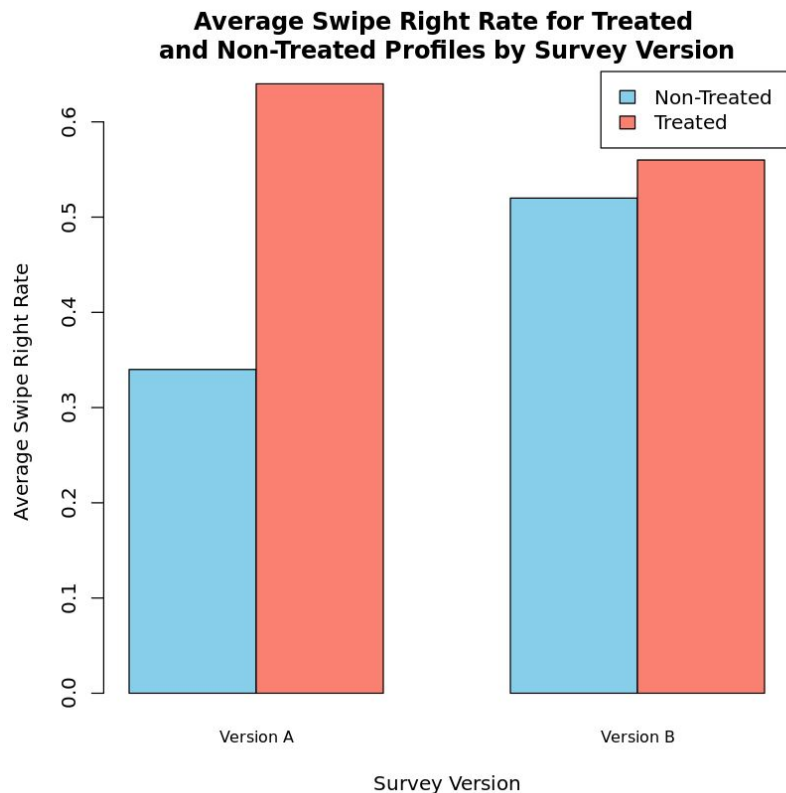
Comparison of Logistic Regression Models for SwipeRight

	Dependent variable:			
	SwipeRight			
	(1)	(2)	(3)	(4)
Treated	0.687** (0.287)	0.689** (0.288)	0.697** (0.289)	0.697** (0.289)
SurveyVersion		0.206 (0.288)		0.045 (0.315)
Educationmiddle_school			0.317 (0.504)	0.296 (0.525)
Educationpostgraduate			0.705 (0.517)	0.680 (0.546)
Educationuniversity			0.070 (0.352)	0.068 (0.352)
Educationvocational_technical_college			0.508 (0.508)	0.499 (0.512)
Constant	-0.282 (0.202)	-0.386 (0.249)	-0.478 (0.294)	-0.493 (0.314)
Observations	200	200	200	200
Log Likelihood	-135.633	-135.375	-134.243	-134.233
Akaike Inf. Crit.	275.265	276.750	280.487	282.466

Note:

*p<0.1; **p<0.05; ***p<0.01

Experiment B Logistic Regression



Comparison of Logistic Regression Models for SwipeRight

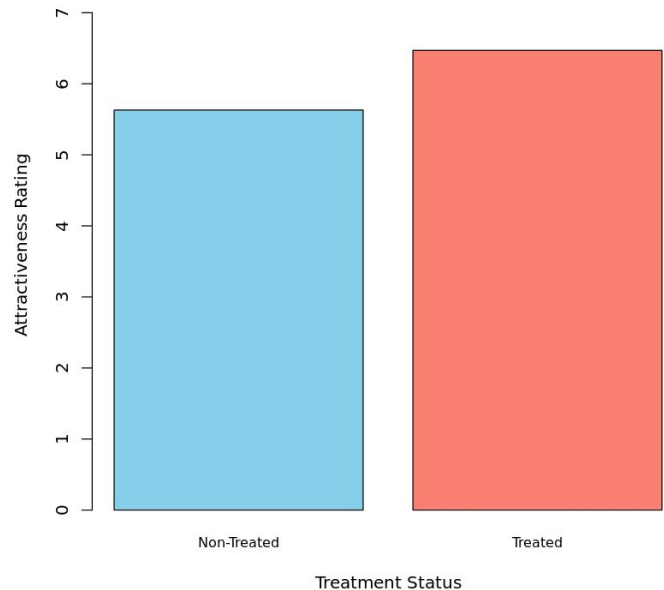
	Dependent variable:			
	SwipeRight			
	(1)	(2)	(3)	(4)
Treated	0.687** (0.287)	0.689** (0.288)	0.697** (0.289)	0.697** (0.289)
SurveyVersion		0.206 (0.288)		0.045 (0.315)
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Akaike Inf. Crit.	275.265	276.750	280.487	282.466

Note:

*p<0.1; **p<0.05; ***p<0.01

Experiment B Linear Regression

Average Attractiveness Rating for Treated and Non-Treated Profiles



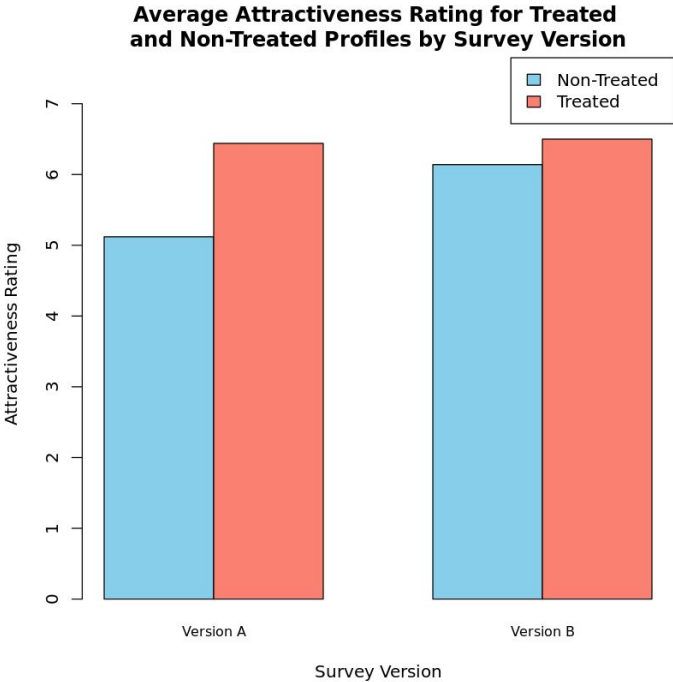
Comparison of Linear Regression Models for AttractivenessRating

	Dependent variable:			
	AttractivenessRating			
	(1)	(2)	(3)	(4)
Treated	0.840** (0.402)	0.840** (0.401)	0.840** (0.397)	0.840** (0.397)
SurveyVersion		0.540 (0.401)		0.372 (0.434)
Educationmiddle_school			-0.588 (0.693)	-0.764 (0.724)
Educationpostgraduate			1.503** (0.693)	1.293* (0.736)
Educationuniversity			-0.163 (0.485)	-0.184 (0.486)
Educationvocational_technical_college			0.912 (0.693)	0.837 (0.699)
Constant	5.630*** (0.284)	5.360*** (0.347)	5.486*** (0.403)	5.358*** (0.430)
Observations	200	200	200	200
R ²	0.022	0.030	0.066	0.070
Adjusted R ²	0.017	0.021	0.042	0.041
Residual Std. Error	2.843 (df = 198)	2.837 (df = 197)	2.805 (df = 194)	2.807 (df = 193)
F Statistic	4.365** (df = 1; 198)	3.097** (df = 2; 197)	2.760** (df = 5; 194)	2.419** (df = 6; 193)

Note:

* p<0.1; ** p<0.05; *** p<0.01

Experiment B Linear Regression

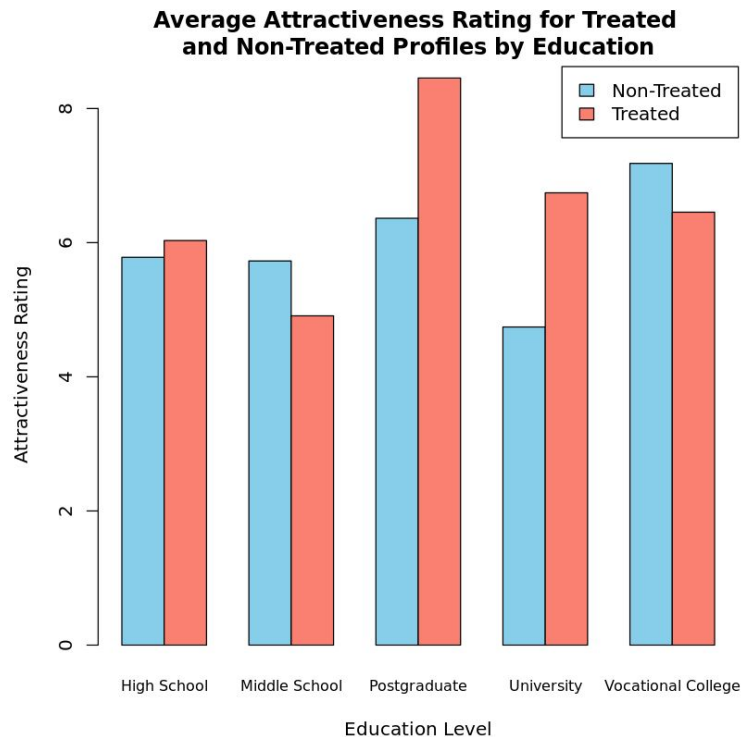


Comparison of Linear Regression Models for AttractivenessRating				
	Dependent variable:			
	AttractivenessRating			
	(1)	(2)	(3)	(4)
Treated	0.840** (0.402)	0.840** (0.401)	0.840** (0.397)	0.840** (0.397)
SurveyVersion		0.540 (0.401)		0.372 (0.434)
Educationmiddle_school			-0.588 (0.693)	-0.764 (0.724)
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Residual Std. Error	2.843 (df = 198)	2.837 (df = 197)	2.805 (df = 194)	2.807 (df = 193)
F Statistic	4.365** (df = 1; 198)	3.097** (df = 2; 197)	2.760** (df = 5; 194)	2.419** (df = 6; 193)

Note:

* p<0.1; ** p<0.05; *** p<0.01

Experiment B Linear Regression



Linear Regression Model with Education Interaction	
	Dependent variable: AttractivenessRating
Treated	0.250 (0.693)
Educationmiddle_school	-0.054 (0.969)
Educationpostgraduate	0.582 (0.969)
Educationuniversity	-1.038 (0.678)
Educationvocational_technical_college	1.401 (0.969)
Treated:Educationmiddle_school	-1.068 (1.370)
Treated:Educationpostgraduate	1.841 (1.370)
Treated:Educationuniversity	1.750* (0.959)
Treated:Educationvocational_technical_college	-0.977 (1.370)
Constant	5.781*** (0.490)
Observations	200
R ²	0.107
Adjusted R ²	0.065
Residual Std. Error	2.773 (df = 190)
F Statistic	2.529*** (df = 9; 190)

Note: *p<0.1; **p<0.05; ***p<0.01

Conclusion

Summary

- While Tinder field experiment proved inconclusive, there was evidence for us to still pursue our initial hypothesis of a treatment effect.
- This hypothesis was confirmed by the results of our Tinder simulation study in which we found a statistically significant positive effect of graduate level education on stated likability of heterosexual women.
- Though these results are promising, we suggest exercising a healthy level of skepticism and encourage further research in this area due to the simulated nature of our study.

Future Work

- Future research can explore the areas of meta-analysis by combining the methodologies of both our field experiment and simulated study to leverage the pros of both approaches.
- This weighted result would capture both the “naturalness” of a field experiment as well as the covariate-controlling-ability of a simulated experiment.
- Such an analysis could weight these results to provide a more robust estimate on the average treatment effect of graduate level education.

THANK YOU
Questions?